AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

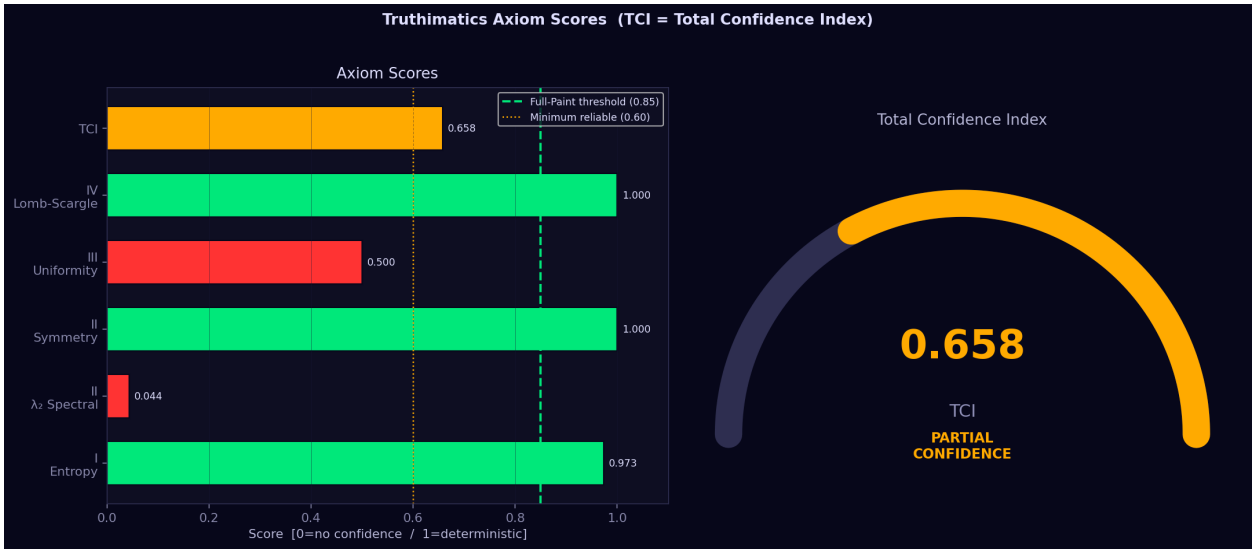
Specimen ID	AXIOM_FCAA83A322
Taxa	Compsognathus
Clade	Theropod
PBDB Records	4
Mass (regression)	2.4 kg (Campione & Evans 2012)
Mass (volumetric)	2.7 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	0.003 m³
Validity Score	1.0000 / 1.0000
TCI	0.6576 (partial)
Axiom I Entropy	0.9734
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.5000
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.012, 0.031, 0.0]
Generated	2026-04-30 23:54 UTC

Ratio Violations

• metatarsal/tibia=0.765 outside [0.28,0.65]

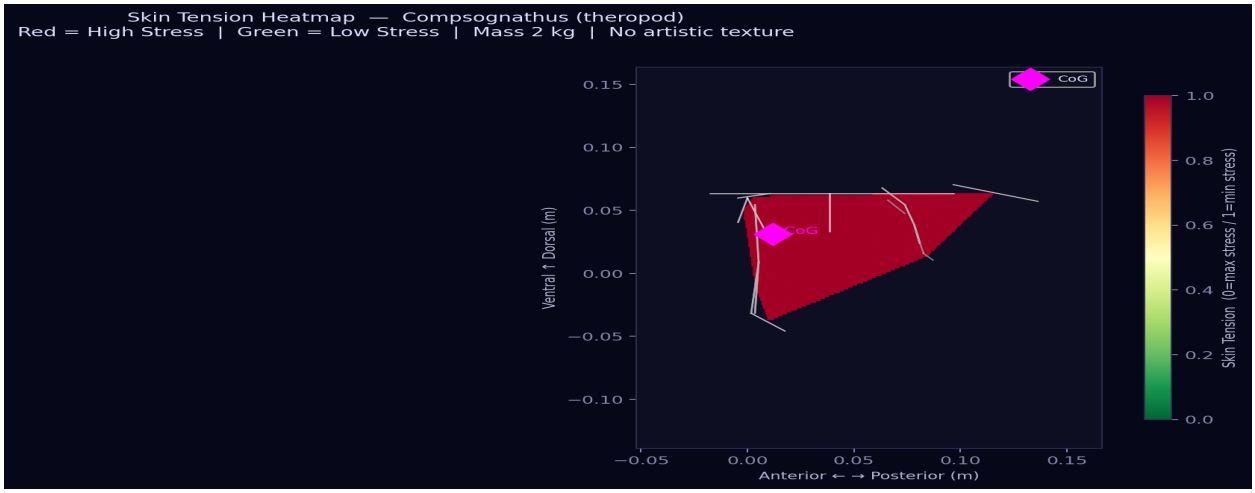
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



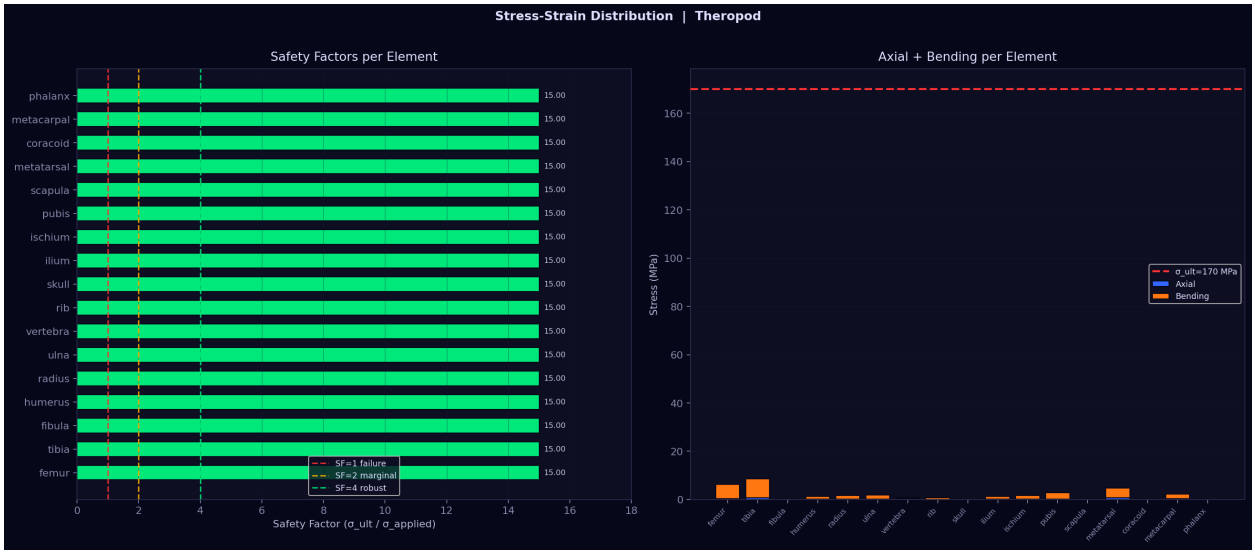
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

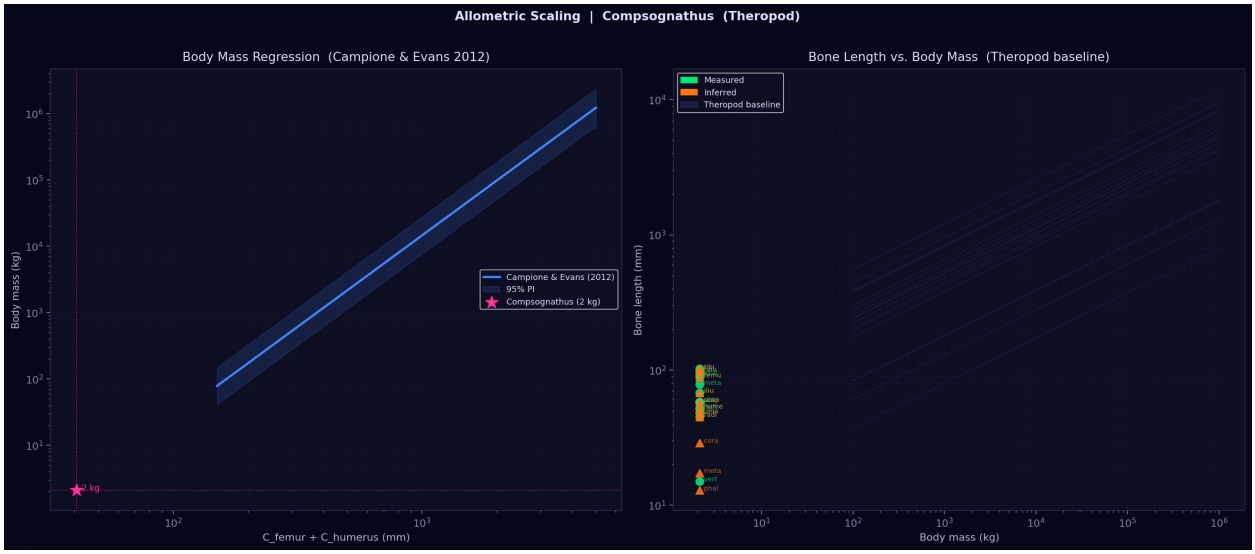


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.005	0.017	0.022	15.000	OK
femur	0.402	5.933	6.335	15.000	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.082	1.149	1.231	15.000	OK
ilium	0.179	1.087	1.265	15.000	OK
ischium	0.184	1.465	1.649	15.000	OK
metacarpal	0.279	2.027	2.305	15.000	OK
metatarsal	0.714	3.989	4.703	15.000	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	0.250	2.595	2.845	15.000	OK
radius	0.080	1.611	1.691	15.000	OK
rib	0.024	0.741	0.765	15.000	OK
scapula	0.002	0.010	0.012	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	0.714	7.824	8.538	15.000	OK
ulna	0.080	1.718	1.798	15.000	OK
vertebra	0.257	0.518	0.775	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.268 m	Femur + tibia + metatarsal
Estimated Speed	4.6 km/h (1.27 m/s)	Gait: Walk | Fr=0.617
Stride Length	0.532 m	Alexander (1976) / Thulborn (1990)
Bite Force	4.6 N (0.5 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	92.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.2856	Circumference / length (dimensionless)
Limb Strength Index	2.2e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	0.87 m	Allometric from femur length
Est. Neck Length	0.12 m	Allometric from femur length
Est. Tail Length	0.33 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	28.9	11.7	1.60	2.0	Constraint	0.40	15.00
femur	88.0	8.0	2.26	2.8	Measured	0.88	15.00
femur_right	88.0	8.0	2.26	2.8	Symmetry	0.84	15.00
fibula	98.0	2.0	0.11	0.1	Measured	0.88	15.00
fibula_right	98.0	2.0	0.11	0.1	Symmetry	0.84	15.00
humerus	52.0	5.0	0.52	0.7	Measured	0.88	15.00
humerus_right	52.0	5.0	0.52	0.7	Symmetry	0.84	15.00
ilium	68.0	12.0	3.92	4.9	Measured	0.88	15.00
ilium_right	68.0	12.0	3.92	4.9	Symmetry	0.84	15.00
ischium	52.0	7.0	1.02	1.3	Measured	0.88	15.00
metacarpal	17.3	2.6	0.02	0.0	Constraint	0.65	15.00
metatarsal	78.0	6.0	1.12	1.4	Measured	0.88	15.00
phalanx	12.8	2.4	0.01	0.0	Constraint	0.40	15.00
pubis	58.0	6.0	0.84	1.0	Measured	0.88	15.00
radius	45.0	3.0	0.16	0.2	Measured	0.88	15.00
radius_right	45.0	3.0	0.16	0.2	Symmetry	0.84	15.00
rib	58.0	3.0	0.21	0.3	Measured	0.88	15.00
scapula	58.0	18.0	7.53	9.4	Measured	0.88	15.00
scapula_right	58.0	18.0	7.53	9.4	Symmetry	0.84	15.00
skull	92.0	45.0	74.62	93.2	Measured	0.88	15.00
tibia	102.0	6.0	1.47	1.8	Measured	0.88	15.00
tibia_right	102.0	6.0	1.47	1.8	Symmetry	0.84	15.00
ulna	48.0	3.0	0.17	0.2	Measured	0.88	15.00
ulna_right	48.0	3.0	0.17	0.2	Symmetry	0.84	15.00
vertebra	15.0	10.0	0.60	0.8	Measured	0.88	15.00